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Analysing Teaching Materials for Interdisciplinary Technology Education Through a Didactic Model

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Abstract

Interdisciplinary STEM education is increasingly promoted in compulsory schooling, yet the role of the technology subject within such approaches remains unevenly articulated. This study examines how technology is positioned in pre-designed teaching materials intended to support interdisciplinary instruction in Swedish lower and upper secondary education. Fourteen materials were analysed using a qualitative, theory-informed content analysis guided by a didactic framework conceptualising teaching practices in terms of what is taught, how teaching is organised, and why activities are framed as educationally meaningful. Through iterative comparison across materials, three recurring configurations of technology emerged: technology-centered, thematically embedded, and tool-oriented projects. While many materials emphasise societal relevance, sustainability, and student engagement, explicit opportunities for students to develop technological knowledge and core design-related practices are often limited. Technology is frequently positioned as contextual support for scientific or societal themes rather than as a distinct epistemic domain. The findings highlight tensions between interdisciplinary ambitions and the disciplinary visibility of technology education and contribute a conceptual framework for understanding how technology is represented in interdisciplinary STEM contexts.

Key Words: Technology Education, STEM, Interdisciplinary Teaching, Didactic Model, Teaching Materials

1 INTRODUCTION

Interdisciplinary STEM education has become an increasingly prominent feature of compulsory schooling, often promoted to address complex societal challenges and to support student engagement through authentic, real-world contexts (Chaojing, 2023). In such teaching designs, technology frequently plays a central yet ambiguous role. While technological solutions, artefacts, and systems are commonly invoked in interdisciplinary themes related to sustainability, energy, or future societies, it is less clear how technology as a school subject is represented and made visible to students within these contexts.

In the Swedish compulsory school system, technology is regularly taught together with science and other subjects, often through thematic or interdisciplinary projects. These arrangements are shaped by curricular intentions as well as by organisational conditions, such as shared teaching responsibilities across subjects. As a result, technology education is often embedded in broader instructional designs rather than taught as a standalone subject. This creates both opportunities and tensions: opportunities for meaningful integration, but also a risk that technology becomes subordinated to other disciplinary perspectives or reduced to a supporting function within interdisciplinary work.

Previous research has shown that when technology is included in interdisciplinary projects, its role can vary considerably. Technology may be foregrounded as a distinct area of knowledge and practice, embedded as contextual content within a broader theme, or rendered largely invisible when technological activities are used primarily to support learning in other subjects. These different configurations shape what kinds of technological knowledge and abilities students are given opportunities to develop. Importantly, such variation has been linked to how technology is understood and positioned by teachers within interdisciplinary teaching designs.

While existing studies have primarily examined teachers' perspectives and instructional practices, less attention has been paid to the role of teaching materials in shaping how technology is articulated within interdisciplinary education. Teaching materials constitute an important interface between curricular intentions and classroom practice. They frame purposes, suggest activities, and implicitly communicate what counts as relevant knowledge within a given teaching design. In interdisciplinary contexts, teaching materials may therefore play a decisive role in how technology is positioned in relation to other subjects and how visible its disciplinary contribution becomes for students.

The didactic model applied in this study builds on a tradition of didactic theorising that conceptualises teaching through three interrelated dimensions: what is taught and learned, how teaching is carried out, and why educational activities are undertaken (Ongstad, 2006; Jank & Meyer, 1997; Dudas, 2019). These dimensions function as analytical questions that make it possible to examine how subject content, teaching practices, and educational purposes are articulated and related in instructional designs.

The model was developed in a previous empirical study of teachers' practices in interdisciplinary STEM education, where it was used to analyse how technology emerges within cross-curricular projects. The results showed that the role of technology varies considerably depending on how these three dimensions are combined in practice. Based on this variation, three recurring configurations were identified: technology-centered, thematically embedded, and tool-oriented projects.

In technology-centered configurations, technology appears as a distinct subject domain with its own content and practices. Here, students are given opportunities to engage in technological processes such as design, problem-solving, and the development of solutions, and technology shapes both the content and the organisation of the teaching. In contrast, thematically embedded configurations position technology within broader interdisciplinary themes, such as sustainability

or innovation. In these cases, technology is present but primarily as contextual content, often linked to scientific explanations or societal issues, with limited emphasis on technological ways of knowing or working. In tool-oriented configurations, technology is further marginalised and mainly functions as a means for learning in other subjects. Technological tools or activities are used to support scientific or mathematical investigations, while technological knowledge and processes remain implicit or underdeveloped.

These configurations illustrate how the visibility and integrity of the technology subject shift across interdisciplinary teaching practices. While all configurations may involve technological artefacts or activities, they differ in the extent to which students are offered opportunities to develop technological understanding as a specific form of knowledge. From a social realist perspective, this can be understood as differences in how knowledge is organised and given status within educational practices, where certain configurations foreground technological knowledge while others subordinate it to other disciplinary aims.

In the present study, this model is used as an analytical lens for examining teaching materials. The three didactic dimensions guide the analysis of how technology is represented in terms of content, activities, and purposes, while the three configurations serve as interpretative tools for identifying patterns in how technology emerges. In this way, the model enables an analysis of not only whether technology is included, but how it is positioned and what kinds of learning opportunities it affords within interdisciplinary contexts.

Against the background of increasing emphasis on interdisciplinary STEM education and ongoing concerns about the disciplinary visibility of technology in such contexts, this study investigates how technology education is epistemically positioned in pre-designed interdisciplinary STEM teaching materials. By analysing how technological knowledge, processes, and purposes are articulated in instructional resources, the study aims to contribute to a conceptual understanding of how disciplinary visibility and integrity are negotiated in interdisciplinary instructional designs. The research question guiding the study is:

> How does the technology subject emerge in pre-designed teaching materials intended for interdisciplinary instruction?

2 LITERATURE REVIEW

Research suggests that interdisciplinary approaches are preferable when the aim is to support students' development of meaningful connections and authentic forms of knowledge (You, 2017; Kumar, 2025). By situating subject content within broader interdisciplinary contexts, students are better able to understand their knowledge and abilities in relation to real-world phenomena (Dyachenko, Mukanova, & Erkibayeva, 2024; Ye & Xu, 2023) which in turn has been shown to increase students' interest in the subject (Widjaja, Hubber, & Aranda, 2019; Michelsen, & Sriraman, 2009; Tonnetti, & Lentillon-Kaestner, 2023).

However, interdisciplinary themes that are taught across several subjects require carefully considered purposes and learning activities. The content, aims, and knowledge bases of the

participating subjects need to be made visible to students, a process that has been shown to be challenging, and therefore requires the formulation of specific learning objectives that make all participating subjects explicit (Cassidy & Puttick, 2022; Chaojing, 2023). To realise the ambitions of interdisciplinary teaching, it is important to work systematically with how interdisciplinary learning is described and how the contributions of different subjects within a theme are articulated (Stevens et al., 2005; Cassidy & Puttick, 2022). Research indicates that certain subjects tend to be marginalised in interdisciplinary collaborations (Yang, Luo, Seah, & van Driel, 2025).

Technology education is particularly at risk of being represented in highly variable ways, depending on how the subject is interpreted by the teacher and positioned within the teaching context (Nordlöf, Hallström, & Höst, 2019). Unlike subjects such as chemistry, physics, or mathematics, technology is a broad subject and cannot be understood as a simplified version of an academic discipline. Previous research has demonstrated that technology education is interpreted in markedly different ways depending on teachers' professional backgrounds, for example whether they are trained as craft teachers, physics teachers, or engineers with teacher education (Bjurulf, 2008; Fahrman, Gumaelius & Norström, 2015). When teachers with different professional backgrounds collaborate in interdisciplinary teaching, or when individual teachers are responsible for implementing interdisciplinary themes, the representation of the technology subject is therefore likely to vary. Similarly, teachers may interpret and enact pre-designed teaching materials in different ways (Kanmaz, 2022).

There are numerous ready-made teaching materials intended to support interdisciplinary instruction that teachers can readily adopt in their practice (Wilcox, Shoulders, & Myers, 2014). These materials typically centre on a contemporary theme and aim to assist teachers in efficiently teaching multiple subjects within a single instructional design. Research indicates that teachers are inclined to use pre-designed teaching materials, but that they relate to and engage with these materials in different ways (Andree & Hansson, 2022). This makes it particularly relevant to examine how subjects are presented within such materials, and how individual subjects are foregrounded or marginalised.

Interdisciplinary education is commonly defined as an approach that involves two or more disciplines and requires explicit integration to address questions or problems that cannot be adequately handled within a single disciplinary perspective. Recent syntheses highlight an emerging consensus that interdisciplinarity is not achieved through thematic alignment alone, but through the integration of disciplinary knowledge and modes of thinking (Boix Mansilla, 2005; see also RTI International, 2022).

Several scholars have further conceptualised interdisciplinarity as part of a continuum ranging from disciplinary and multidisciplinary approaches to interdisciplinary and transdisciplinary ones (Drake & Burns, 2004, Klein, 2006). Within this continuum, multidisciplinary or cross-disciplinary approaches juxtapose subjects without requiring synthesis, whereas interdisciplinary approaches are characterised by explicit integration of disciplinary perspectives. This distinction is particularly relevant when analysing how technology is positioned in interdisciplinary STEM materials, as thematic coherence does not necessarily imply epistemic integration.

Instructional materials research highlights how teaching resources shape both curriculum enactment and disciplinary meaning-making (Remillard, 2005; Davis & Krajcik, 2005). Pre-designed materials not only structure teaching activities but also communicate implicit assumptions about knowledge, pedagogy, and educational purposes. In interdisciplinary contexts, such materials often provide ready-made thematic or project-based designs, thereby influencing how disciplinary connections are articulated and how subject-specific epistemic practices are foregrounded or marginalised. Analysing pre-designed materials therefore offers insight into how technology education is positioned within broader interdisciplinary STEM teaching.

3 METHODOLOGY

This study employed a qualitative, theory-informed content analysis of pre-designed teaching materials intended to support interdisciplinary STEM projects in Swedish lower and upper secondary education. Pre-designed materials are understood as instructional resources developed outside individual classrooms (e.g. by educational organisations, public agencies, museums, NGOs, or educational initiatives) and made available for teachers to use, adapt, or implement. Such materials constitute mediators between curriculum intentions and classroom practice (Remillard, 2005; Davis & Krajcik, 2005).

Selection of materials followed purposive sampling aimed at analytical variation rather than representativeness. Materials were included if they (1) explicitly addressed interdisciplinary themes involving technology and at least one other subject, (2) were intended for Swedish compulsory schooling, and (3) provided structured teaching guidance such as lesson plans or project descriptions. Fourteen materials meeting these criteria were selected from widely used national repositories and portals. The sampling aimed to capture variation in thematic orientation, institutional origin, and intended educational level, in order to explore different ways in which the technology subject is positioned across interdisciplinary teaching materials.

The analysis was guided by a didactic framework conceptualising teaching practices along three dimensions: what is taught and learned, how teaching is organised and enacted, and why activities are framed as educationally meaningful (Lennholm & Engström, 2025). This framework was used as an analytical lens to examine how technological knowledge, activities, and purposes were articulated across the material set.

The analytical procedure followed an iterative process inspired by qualitative structuring content analysis (Mayring, 2014). Materials were first read holistically, then segments related to technological content, activities, and purposes were coded in relation to the analytical dimensions. Through iterative comparison across materials, three recurring configurations were identified: technology-centered, thematically embedded, and tool-oriented projects. These configurations were empirically derived and not predetermined by the framework.

Analytical credibility was strengthened through iterative comparison, joint interpretation by the authors, and refinement of category boundaries to ensure consistency in the identification of patterns. Category stability was assessed through repeated comparison across the dataset, and interpretations were refined through collaborative analytical discussions between the authors.

Importantly, the analytical framework did not predetermine the configurations presented in the results. Rather, the dimensions of what, how, and why functioned as an interpretive lens guiding the examination of how technological knowledge, activities, and purposes were articulated across the material set. Through iterative comparison and synthesis, three recurring patterns emerged describing different ways in which the technology subject was positioned within interdisciplinary teaching designs. These configurations therefore represent empirically derived analytical constructs rather than direct extensions of the framework's dimensions. The *why* dimension was operationalised through explicit statements of educational purpose, societal framing, or intended competencies articulated in the materials, such as references to sustainability, innovation, or student engagement.

A generative artificial intelligence (GenAI) tool (ChatGPT, OpenAI) was used as a supportive tool in the writing process, including for language editing, structuring of sections, and refinement of arguments. The tool was also used for discussing ideas and improving clarity of presentation. It was not used to generate empirical data, conduct analysis, or produce substantive research findings. All research design, analysis, interpretation, and final decisions regarding the manuscript were undertaken by the authors, who bear full responsibility for its content.

4 RESULTS

In total, 14 lesson plans/teaching materials were analysed. All of these are provided by well-known and well-established actors, most of whom are not primarily oriented towards the school sector but have developed educational programmes as part of their decision to also contribute to school education. In the results section below, we describe for each teaching material how the content, knowledge, and abilities of the technology subject are presented, addressed, and made visible. This is discussed in relation to the model described above as well as through other themes that emerge from the analysis. The analysis of commonly used teaching materials for interdisciplinary STEM projects in Swedish lower and upper secondary education reveals three distinct configurations of how the technology subject is positioned: technology-centered, thematically embedded, and tool-oriented projects. Across these configurations, recurring patterns emerge regarding what aspects of technological knowledge are foregrounded, how activities are structured, and why projects are framed as educationally meaningful.

The configurations describe recurring patterns in how technology is epistemically positioned within interdisciplinary instructional designs, rather than representing distinct pedagogical models or mutually exclusive categories. The analysis revealed systematic variation in how the technology subject was articulated and positioned within interdisciplinary STEM teaching materials. While all materials addressed interdisciplinary themes and emphasised societal relevance, they differed in the extent to which technological knowledge, processes, and evaluative practices were made explicit. Across the material set, technology was framed either as a primary object of learning, as contextual content within broader thematic narratives, or as a practical means of supporting learning in other subjects. Through comparative analysis, three analytically distinct configurations were identified: technology-centered, thematically embedded, and tool-oriented projects. These configurations represent dominant tendencies rather than mutually exclusive categories.

Table 1 illustrates how some commonly used Swedish repositories align with the three identified configurations. While all materials emphasise relevance and engagement, explicit opportunities for technological inquiry vary considerably across configurations. Overall, the distribution of examples across configurations indicates a predominance of thematically embedded approaches, with comparatively fewer technology-centered designs. This pattern suggests that while interdisciplinary ambitions are strongly represented in the materials, explicit articulation of technological knowledge and design-related reasoning is less consistently foregrounded. The findings therefore point to a tension between thematic integration and disciplinary visibility, where the organisational logic of interdisciplinary projects may shape how technological knowledge is framed as an object of learning.

Table 1

Examples of Swedish teaching materials mapped to configurations of technology in interdisciplinary STEM projects

Repository / Example (link)	**Dominant what**	**Dominant how**	**Dominant why**	**Typical omission / tension**
Technology- centered				
1 CETIS – Teknik tillsammans https://liu.se/cetis	Technology concepts + tech-society relations	Structured work areas	Strengthen technology as a school subject	Iterative design, evaluation criteria, systems thinking under-developed
2 NTA – Smartare produkter (micro:bit) https://www.nta.se	Technical problem-solving, programming	Practical construction & testing	Capability & interest in technology	Design often linear; limited optimisation reasoning
3 Komtek – municipal tech workshops https://komtek.se/	Hands-on technology making	Workshop-oriented activities	Build confidence & practical skills	Technology framed as "making" rather than reasoning
Thematically embedded				
4 Naturskyddsföreningen – school materials https://www.naturskyddsforeningen.se/skola/skolmaterial/	Energy, climate, society	Theme-based interdisciplinary lessons	Sustainability & citizenship	Technology mainly illustrative
5 NTA – Energi och hållbar utveckling https://www.nta.se	Societal challenges	Student participation in SSI	Engagement & relevance	Tech knowledge progression unclear

6 Tekniska museet – Zero City https://www.tekniskamuseet.se/skola/stem-skolprojekt-som-inspirerar/	Future cities & sustainability	Pre/during/after visit activities	Motivation & future orientation	Tech processes depend on teacher -follow-up
7 Tekniska museet – Hyper Human (self-guided visit) https://www.tekniskamuseet.se/skola/skolbesok-pa-egen-hand/hyper-human/	Human–technology relations	Exhibition-based exploration	Engagement with tech & society	Minimal scaffolding of tech reasoning
8 Unga Innovatörer – Fossilfrihet challenge https://ungainnovatorer.se/utmaningar/fossilfrihet	Innovation linked to climate action	Challenge-based project work	Relevance, innovation, action	May remain thematic without deep tech processes
9 Globala målen – school materials https://globalamalen.se/skola/skolmaterial/	UN SDGs across subjects	Cross-curricular lessons	Sustainability & global awareness	Technology not treated as distinct inquiry
10 Skogen i skolan – "Mitt liv som träfiber" https://www.skogeniskolan.se/laromedel/mitt-liv-som-trafiber-lararhandledning/	Bioeconomy, materials	Hands-on + reflection	Systems understanding of forest use	Design/evaluation steps weakly articulated
Tool- oriented				
11 UR – Typiskt teknik https://urplay.se/amne/teknik	Everyday technology	Media-supported short tasks	Accessibility, engagement	Descriptive rather than evaluative
12 Digitala lektioner Teknikhistoria: Gladys & GPS https://digitalalektioner.se/lektion/teknikhistoria-gladys-och-gps-tekniken/	Tech history & navigation	Online lesson + narrative	Understanding technological development	Limited engagement with design reasoning
13 ESERO – Moon Camp Mission https://www.esero.se/moon-camp-mission/	Space technology context	Digital inquiry & modelling	Motivation through real-world STEM	Technology as environment/tool
14 Nobel Prize Museum – school programs (lower & upper secondary) https://www.nobelprizemuseum.se/skola/skolprogram/#hogstadiet	Science & innovation history	Museum programs & activities	Inspiration & recognition of knowledge	Tech inquiry often historical /descriptive

5 DISCUSSION

The findings contribute to discussions in both technology education and interdisciplinary education research concerning the conditions under which disciplinary knowledge becomes visible or diluted in interdisciplinary teaching contexts. While interdisciplinary STEM approaches are often promoted for enhancing relevance and engagement, the analysis suggests that such ambitions do not automatically ensure explicit integration of technological knowledge and reasoning. The predominance of thematically embedded and tool-oriented configurations indicates that instructional designs frequently prioritise thematic coherence over systematic development of technological knowledge. Practically, this highlights the importance of making technological processes such as design iteration, evaluation, and systems reasoning explicit when designing interdisciplinary STEM instruction. Strengthening the epistemic visibility of technology may support more robust learning outcomes and clearer positioning of the subject within interdisciplinary practice.

Technology-centered projects are comparatively rare in the material analysed. When present, they typically express strong ambitions to foreground innovation, future solutions, or technological development. However, these ambitions are only partially realised in practice. While students are often encouraged to generate ideas or propose solutions, opportunities to engage in core technological processes—such as iterative design, evaluation against criteria, optimisation, or systematic comparison of alternatives—are limited. Systems thinking, although implicitly relevant in themes such as sustainability or infrastructure, is seldom articulated as an explicit object of student reasoning. As a result, technology tends to appear primarily as a source of solutions rather than as a discipline characterised by distinct ways of knowing and reasoning.

Thematically embedded projects constitute the dominant configuration across the materials. These projects are organised around broad societal or environmental themes—such as climate change, sustainable development, or future cities—and integrate content from science, social studies, and technology. Within this configuration, interdisciplinarity is primarily achieved at the thematic level rather than through epistemic integration of disciplinary perspectives.

Technology is frequently positioned in an illustrative or supportive role, providing examples of existing solutions or applications within a broader narrative largely shaped by scientific explanations or societal considerations. While the materials emphasise values such as sustainability, collaboration, and student engagement, they rarely specify how technological understanding is expected to develop over time or how it differs from scientific or social knowledge. Consequently, technology's disciplinary contribution risks being diluted, as thematic coherence takes precedence over conceptual clarity within the technology subject.

In tool-oriented projects, technology is primarily framed as a means for learning rather than as an object of learning. Digital tools, simulations, programming environments, or prototyping materials are commonly used to support investigations in science or mathematics, or to visualise societal problems. Although such activities may involve building or modelling, the learning objectives are seldom articulated in explicitly technological terms.

Students are rarely required to justify design choices, evaluate solutions using explicit criteria, or reflect on the limitations and consequences of technological decisions. As a result, technological activity tends to become procedural rather than epistemic. While this configuration often supports practical classroom implementation and student engagement, it reinforces a view of technology as an auxiliary or service subject rather than as a field of inquiry with its own knowledge structures and evaluative practices.

Across all three configurations, the materials consistently articulate strong educational purposes related to sustainability, problem-solving, and societal relevance. Similar patterns are described in broader analyses of interdisciplinary education, which point to accountability pressures, organisational constraints, and uneven access to professional support as factors that tend to favour low-risk forms of interdisciplinarity, where disciplinary integration remains implicit (RTI International, 2022). However, these purposes are only weakly connected to explicit technological knowledge development. Technology is frequently mobilised to address why questions—such as why sustainability matters or why innovation is needed—without sufficient attention to what constitutes technological understanding or how such understanding is developed through teaching.

Taken together, the findings suggest that while interdisciplinary STEM materials successfully promote thematic breadth and student engagement, they often underrepresent technology as a distinct epistemic domain. This imbalance has implications for how students encounter technological knowledge and for how teachers navigate the role of the technology subject within interdisciplinary STEM practices.

Across repositories, thematic coherence and motivational purposes are well developed, while explicit opportunities for technological reasoning remain limited. Even in materials where technology appears central, technological processes such as evaluation, iteration, and systems thinking are weakly articulated. The configuration of technology strongly shapes what kinds of knowledge students are invited to develop.

This study contributes to technology education research by conceptualising how instructional materials mediate the disciplinary visibility of technology within interdisciplinary STEM contexts. By extending interdisciplinary education research through a technology-specific analytical lens, the study highlights the importance of epistemic clarity when designing interdisciplinary learning environments. It also demonstrates how teaching materials function as key mediators between curriculum intentions and classroom practice, shaping the conditions under which technological knowledge is recognised as a distinct domain of learning.

Taken together, the findings suggest that strengthening the epistemic visibility of technology is essential if interdisciplinary STEM education is to support both thematic relevance and disciplinary learning progression.

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